Litmus: KV-Cache Reuse and the Accuracy-Latency Frontier of On-Device Manipulation Detection

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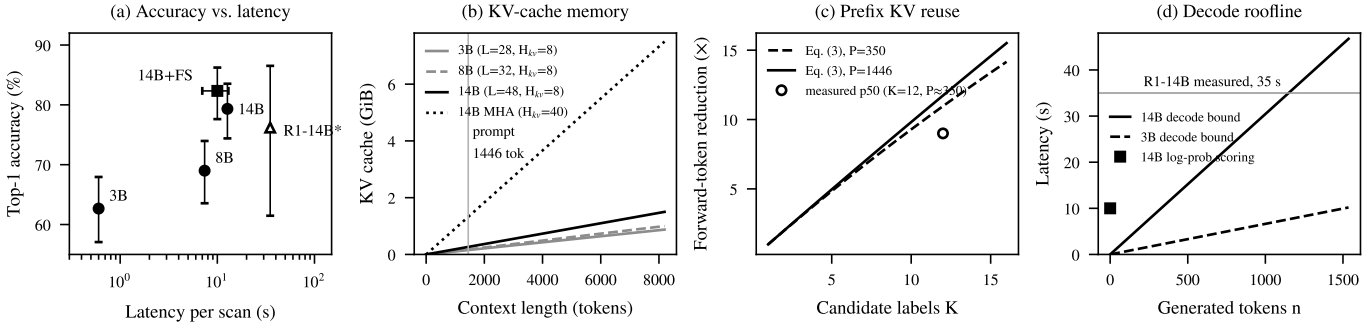


Fig. 1: (a) Accuracy vs. latency on Apple M4 Pro, Wilson 95% intervals (*R1: balanced N=42 subset). (b) fp16 KV cache from each `config.json`; dotted is a 14B multi-head counterfactual. (c) Eq. (3) vs. measured p50 speedup. (d) Eq. (4) decode bound.

Abstract: **Litmus labels a text selection with one of 14 persuasion techniques or *none* by scoring candidate labels under a local 4-bit LLM. This note analyzes KV-cache memory, prefix KV reuse, and the decode bandwidth bound. Held-out top-1 accuracy rose from 62.7% (3B) to 82.3% (14B, few-shot); prefix reuse cut median latency 9.0×.**

I. Scoring as Constrained Decoding

The detector never generates text. For input x and each label $y \in \mathcal{Y}$ (K=$|\mathcal{Y}|$=15) it computes

$$s(y \mid x) = \frac{1}{|y|} \sum_{t=1}^{|y|} \log p_\theta(y_t \mid x, y_{<t}) \tag{1}$$

and returns $\arg\max_y s$. Length normalization removes the bias toward short labels (1 to 7 tokens). A two-label gate first screens benign text (PR-AUC 0.93). Ranked scores give top-2 accuracy of 73.0% against 64.0% top-1 on the 3B, so most errors name a co-present technique.

II. KV-Cache Memory

Each attention layer caches one key and one value per token:

$$M_{\text{tok}} = 2\, L\, H_{kv}\, d_h\, b. \tag{2}$$

All three models use grouped-query attention [1] with H_{kv}=8, d_h=128, giving 112, 128 and 192 KiB per token (3B, 8B, 14B, fp16). The 1446-token few-shot prompt occupies 271 MiB on the 14B, against 1.32 GiB under full multi-head attention with H_{kv}=40 (Fig. 1b). Beside 8.31 GB of weights the cache is minor, so RAM did not limit the 14B on a 24 GB machine.

III. Prefix KV Reuse

All labels share a P-token prefix. The deployed scorer prefills it once, appends each label to the cache, reads its log-probabilities, then trims the cache back to length P. Verdicts are unchanged. Relative to K independent passes, forwarded tokens fall by

$$S(K) = \frac{K(P + \bar{\ell})}{P + K(\bar{\ell} - 1)}. \tag{3}$$

At P=1446 the scorer forwards 1493 tokens instead of 21,752 (14.6×). At measurement time (K=12, P≈350) Eq. (3) predicted 11.0×; median latency fell from 4014 to 446 ms (9.0×, Fig. 1c). The gap is fixed per-call overhead and label tokens attending over the full prefix.

IV. Prefill Versus Decode

Prefill is parallel and compute-bound. Decode emits one token per step and streams all weights each step:

$$t_{\text{dec}}(n) \;\geq\; nW/B. \tag{4}$$

With W=8.31 GB and B=273 GB/s, the 14B needs at least 30.4 ms per token. Log-prob scoring decodes zero tokens. DeepSeek-R1-Distill-14B [2] must decode a chain of thought first: it averaged 35 s per scan, which bounds its mean chain at about 1150 tokens (Fig. 1d), and reached only 76.2% on a balanced subset.

V. Results and Discussion

Accuracy rose monotonically with size (Table I), so 62.7% was a 3B capacity limit. Few-shot exemplars helped the 14B by three points but not the 3B. LoRA [3] on 503 imbalanced examples did not beat few-shot, and self-consistency voting [4] gained nothing on cleaned labels. The 3B serves the 0.6 s shortcut and the 14B is an opt-in deep scan: the split is set by compute and bandwidth, not memory.

References

[1] J. Ainslie *et al.*, "GQA: Training generalized multi-query transformer models from multi-head checkpoints," in *Proc. EMNLP*, 2023.
[2] DeepSeek-AI, "DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning," arXiv:2501.12948, 2025.
[3] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
[4] X. Wang *et al.*, "Self-consistency improves chain of thought reasoning in language models," in *Proc. ICLR*, 2023.

TABLE I: Held-out benchmark (N=300), all on-device

Configuration	Top-1 (%)	MCC	Latency (s)
3B, few-shot (shipped)	62.7	0.55	0.6
+ gate fix, clean set	66.9	n/a	0.6
8B, zero-shot	69.0	0.634	7.4
14B, zero-shot	79.3	0.752	12.7
14B, few-shot	**82.3**	**0.787**	7 to 13
14B, LoRA (N=123)	78.9	n/a	n/a